

Math 151A Homework 6
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Problem 1: Section 4.1, Question 6

Use the most accurate three-point formula to approximate $f'(x)$ at each point.
For equally spaced x -values with step size h :

$$\text{Forward 3-point: } f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h},$$

$$\text{Central 3-point: } f'(x_0) \approx \frac{f(x_0 + h) - f(x_0 - h)}{2h},$$

$$\text{Backward 3-point: } f'(x_n) \approx \frac{3f(x_n) - 4f(x_n - h) + f(x_n - 2h)}{2h}.$$

(a)

Points: $x = -0.3, -0.2, -0.1, 0$ with $h = 0.1$.

$$\begin{aligned} f'(-0.3) &\approx \frac{-3f(-0.3) + 4f(-0.2) - f(-0.1)}{2(0.1)} = \frac{-3(-0.27652) + 4(-0.25074) - (-0.16134)}{0.2} \\ &\approx -0.0603. \end{aligned}$$

$$f'(-0.2) \approx \frac{f(-0.1) - f(-0.3)}{2(0.1)} = \frac{-0.16134 - (-0.27652)}{0.2} \approx 0.5759.$$

$$f'(-0.1) \approx \frac{f(0) - f(-0.2)}{2(0.1)} = \frac{0 - (-0.25074)}{0.2} \approx 1.2537.$$

$$f'(0) \approx \frac{3f(0) - 4f(-0.1) + f(-0.2)}{2(0.1)} = \frac{3 \cdot 0 - 4(-0.16134) + (-0.25074)}{0.2} \approx 1.9731.$$

(b)

Points: $x = 7.4, 7.6, 7.8, 8.0$ with $h = 0.2$.

$$f'(7.4) \approx \frac{-3f(7.4) + 4f(7.6) - f(7.8)}{2(0.2)} = \frac{-3(-68.3193) + 4(-71.6982) - (-75.1576)}{0.4} \approx -16.6933,$$

$$f'(7.6) \approx \frac{f(7.8) - f(7.4)}{2(0.2)} = \frac{-75.1576 - (-68.3193)}{0.4} \approx -17.0958,$$

$$f'(7.8) \approx \frac{f(8.0) - f(7.6)}{2(0.2)} = \frac{-78.6974 - (-71.6982)}{0.4} \approx -17.4980,$$

$$f'(8.0) \approx \frac{3f(8.0) - 4f(7.8) + f(7.6)}{2(0.2)} = \frac{3(-78.6974) - 4(-75.1576) + (-71.6982)}{0.4} \approx -17.9000.$$

(c)

Points: $x = 1.1, 1.2, 1.3, 1.4$ with $h = 0.1$.

$$f'(1.1) \approx \frac{-3f(1.1) + 4f(1.2) - f(1.3)}{2(0.1)} = \frac{-3(1.52918) + 4(1.64024) - 1.70470}{0.2} \approx 1.3436,$$

$$f'(1.2) \approx \frac{f(1.3) - f(1.1)}{2(0.1)} = \frac{1.70470 - 1.52918}{0.2} \approx 0.8776,$$

$$f'(1.3) \approx \frac{f(1.4) - f(1.2)}{2(0.1)} = \frac{1.71277 - 1.64024}{0.2} \approx 0.36265,$$

$$f'(1.4) \approx \frac{3f(1.4) - 4f(1.3) + f(1.2)}{2(0.1)} = \frac{3(1.71277) - 4(1.70470) + 1.64024}{0.2} \approx -0.20125.$$

(d)

Points: $x = -2.7, -2.5, -2.3, -2.1$ with $h = 0.2$.

$$f'(-2.7) \approx \frac{-3f(-2.7) + 4f(-2.5) - f(-2.3)}{2(0.2)} = \frac{-3(0.054797) + 4(0.11342) - 0.65536}{0.4} \approx -0.91518,$$

$$f'(-2.5) \approx \frac{f(-2.3) - f(-2.7)}{2(0.2)} = \frac{0.65536 - 0.054797}{0.4} \approx 1.50141,$$

$$f'(-2.3) \approx \frac{f(-2.1) - f(-2.5)}{2(0.2)} = \frac{0.98472 - 0.11342}{0.4} \approx 2.17825,$$

$$f'(-2.1) \approx \frac{3f(-2.1) - 4f(-2.3) + f(-2.5)}{2(0.2)} = \frac{3(0.98472) - 4(0.65536) + 0.11342}{0.4} \approx 1.11535.$$

Problem 2: Section 4.2, Question 1

We apply Richardson extrapolation (Example 1) to the centered difference

$$D(h) = \frac{f(x_0 + h) - f(x_0 - h)}{2h}$$

using step sizes h , $h/2$, $h/4$.

First level:

$$N_1(h) = D(h), \quad N_1\left(\frac{h}{2}\right) = D\left(\frac{h}{2}\right), \quad N_1\left(\frac{h}{4}\right) = D\left(\frac{h}{4}\right).$$

Second level (remove $O(h^2)$ error):

$$N_2(h) = N_1\left(\frac{h}{2}\right) + \frac{N_1\left(\frac{h}{2}\right) - N_1(h)}{3}, \quad N_2\left(\frac{h}{2}\right) = N_1\left(\frac{h}{4}\right) + \frac{N_1\left(\frac{h}{4}\right) - N_1\left(\frac{h}{2}\right)}{3}.$$

Third level (remove $O(h^4)$ error):

$$N_3(h) = N_2\left(\frac{h}{2}\right) + \frac{N_2\left(\frac{h}{2}\right) - N_2(h)}{15}.$$

We now compute $N_3(h)$ for each function with $h = 0.4$.

(a) $f(x) = \ln x$, $x_0 = 1.0$

$$D(0.4) = \frac{\ln(1.4) - \ln(0.6)}{0.8} \approx 1.059122,$$

$$D(0.2) = \frac{\ln(1.2) - \ln(0.8)}{0.4} \approx 1.013663, \quad D(0.1) = \frac{\ln(1.1) - \ln(0.9)}{0.2} \approx 1.003353.$$

$$N_2(0.4) = D(0.2) + \frac{D(0.2) - D(0.4)}{3} \approx 0.998510,$$

$$N_2(0.2) = D(0.1) + \frac{D(0.1) - D(0.2)}{3} \approx 0.999917.$$

$$N_3(0.4) = N_2(0.2) + \frac{N_2(0.2) - N_2(0.4)}{15} \approx 1.000011 \approx 1.000.$$

So $N_3(h) \approx 1$, which matches $f'(1) = 1/x|_{x=1} = 1$.

(b) $f(x) = x + e^x$, $x_0 = 0$

$$D(0.4) = \frac{f(0.4) - f(-0.4)}{0.8} \approx 2.026881, \quad D(0.2) \approx 2.006680, \quad D(0.1) \approx 2.001668.$$

$$N_2(0.4) = 2.006680 + \frac{2.006680 - 2.026881}{3} \approx 1.999946,$$

$$N_2(0.2) = 2.001668 + \frac{2.001668 - 2.006680}{3} \approx 1.999997,$$

$$N_3(0.4) = 1.999997 + \frac{1.999997 - 1.999946}{15} \approx 2.000000.$$

Thus $N_3(h) \approx 2$, agreeing with $f'(0) = 1 + e^0 = 2$.

(c) $f(x) = 2^x \sin x$, $x_0 = 1.05$

Compute the centered differences:

$$D(0.4) \approx 2.203166, \quad D(0.2) \approx 2.257237, \quad D(0.1) \approx 2.270674.$$

Extrapolate:

$$N_2(0.4) = 2.257237 + \frac{2.257237 - 2.203166}{3} \approx 2.275261,$$

$$N_2(0.2) = 2.270674 + \frac{2.270674 - 2.257237}{3} \approx 2.275153,$$

$$N_3(0.4) = 2.275153 + \frac{2.275153 - 2.275261}{15} \approx 2.275146.$$

So $N_3(h) \approx 2.275146$.

(d) $f(x) = x^3 \cos x$, $x_0 = 2.3$

$$D(0.4) \approx -19.471761, \quad D(0.2) \approx -19.606223, \quad D(0.1) \approx -19.636854.$$

$$N_2(0.4) = -19.606223 + \frac{-19.606223 + 19.471761}{3} \approx -19.651044,$$

$$N_2(0.2) = -19.636854 + \frac{-19.636854 + 19.606223}{3} \approx -19.647064,$$

$$N_3(0.4) = -19.647064 + \frac{-19.647064 + 19.651044}{15} \approx -19.646799.$$

Hence $N_3(h) \approx -19.6468$.

Problem 3: Section 4.3, Question 2

Approximate each definite integral using the (single-interval) Trapezoidal rule

$$\int_a^b f(x) dx \approx T = \frac{b-a}{2} [f(a) + f(b)].$$

(a)

$$\int_{-0.25}^{0.25} (\cos x)^2 dx, \quad f(x) = (\cos x)^2, \quad a = -0.25, \quad b = 0.25.$$
$$T = \frac{0.25 - (-0.25)}{2} [f(-0.25) + f(0.25)] = \frac{0.5}{2} [\cos^2(-0.25) + \cos^2(0.25)].$$

Since $\cos^2(-0.25) = \cos^2(0.25)$,

$$T = 0.25 (2 \cos^2(0.25)) = 0.5 \cos^2(0.25) \approx 0.469396.$$

(b)

$$\int_{-0.5}^0 x \ln(x+1) dx, \quad f(x) = x \ln(x+1), \quad a = -0.5, \quad b = 0.$$
$$T = \frac{0 - (-0.5)}{2} [f(-0.5) + f(0)] = \frac{0.5}{2} [(-0.5) \ln(0.5) + 0 \cdot \ln 1] = 0.25 [-0.5 \ln 0.5] \approx 0.086643.$$

(c)

$$\int_{0.75}^{1.3} ((\sin x)^2 - 2x \sin x + 1) dx,$$
$$f(x) = (\sin x)^2 - 2x \sin x + 1, \quad a = 0.75, \quad b = 1.3.$$
$$T = \frac{1.3 - 0.75}{2} [f(0.75) + f(1.3)] = \frac{0.55}{2} [f(0.75) + f(1.3)].$$

Compute $f(0.75)$ and $f(1.3)$ numerically, then

$$T \approx 0.275 [f(0.75) + f(1.3)] \approx -0.037024.$$

(d)

$$\int_e^{e+1} \frac{1}{x \ln x} dx, \quad f(x) = \frac{1}{x \ln x}, \quad a = e, \quad b = e+1.$$
$$T = \frac{(e+1) - e}{2} [f(e) + f(e+1)] = \frac{1}{2} \left[\frac{1}{e \ln e} + \frac{1}{(e+1) \ln(e+1)} \right].$$

Since $\ln e = 1$,

$$T = \frac{1}{2} \left[\frac{1}{e} + \frac{1}{(e+1) \ln(e+1)} \right] \approx 0.286334.$$

Problem 4: Section 4.5, Question 17

We show that $R_{k,2}$ from Romberg integration is the same as the Composite Simpson rule with step $h = h_k$.

Let $[a, b]$ be divided into n subintervals of equal width $h_k = \frac{b-a}{n}$, where n is even. Denote $x_j = a + jh_k$ for $j = 0, 1, \dots, n$.

Trapezoidal approximations

The composite trapezoidal rule with step h_k (using n subintervals) is

$$T(h_k) = \frac{h_k}{2} \left[f(x_0) + 2 \sum_{j=1}^{n-1} f(x_j) + f(x_n) \right].$$

If we double the step size to $h_{k-1} = 2h_k$, then there are $n/2$ subintervals and the nodes are $x_0, x_2, x_4, \dots, x_n$. The corresponding trapezoidal approximation is

$$T(h_{k-1}) = \frac{h_{k-1}}{2} \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + f(x_n) \right] = h_k \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + f(x_n) \right],$$

since $h_{k-1} = 2h_k$.

Expression for $T(h_k)$

Split the sum in $T(h_k)$ into even and odd indices:

$$\sum_{j=1}^{n-1} f(x_j) = \sum_{j=1}^{n/2-1} f(x_{2j}) + \sum_{j=1}^{n/2} f(x_{2j-1}).$$

So

$$T(h_k) = \frac{h_k}{2} \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + 2 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right].$$

Romberg entry $R_{k,2}$

Romberg integration defines

$$R_{k,1} = T(h_k), \quad R_{k-1,1} = T(h_{k-1}),$$

and

$$R_{k,2} = R_{k,1} + \frac{R_{k,1} - R_{k-1,1}}{4 - 1} = \frac{4T(h_k) - T(h_{k-1})}{3}.$$

Substitute the formulas for $T(h_k)$ and $T(h_{k-1})$:

$$4T(h_k) = 2h_k \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + 2 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right],$$

$$T(h_{k-1}) = h_k \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + f(x_n) \right].$$

Therefore,

$$\begin{aligned} 4T(h_k) - T(h_{k-1}) &= h_k \left[2f(x_0) + 4 \sum_{j=1}^{n/2-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + 2f(x_n) \right. \\ &\quad \left. - f(x_0) - 2 \sum_{j=1}^{n/2-1} f(x_{2j}) - f(x_n) \right] \\ &= h_k \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right]. \end{aligned}$$

Divide by 3:

$$R_{k,2} = \frac{4T(h_k) - T(h_{k-1})}{3} = \frac{h_k}{3} \left[f(x_0) + 2 \sum_{j=1}^{n/2-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right].$$

□